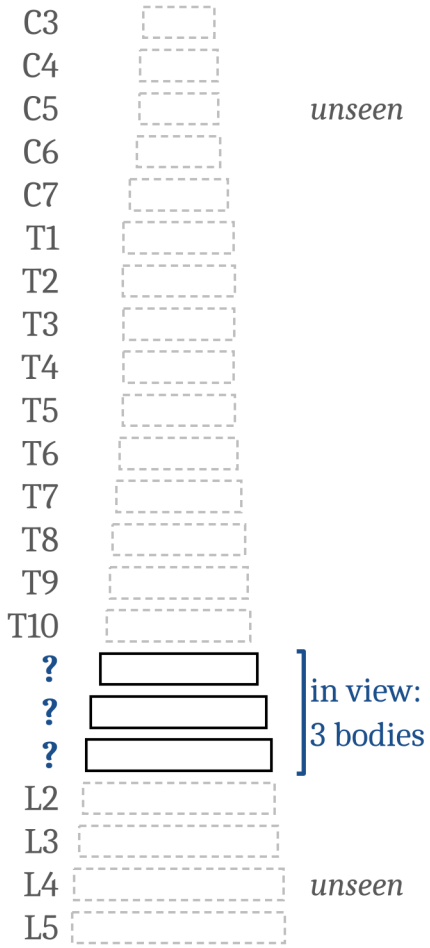
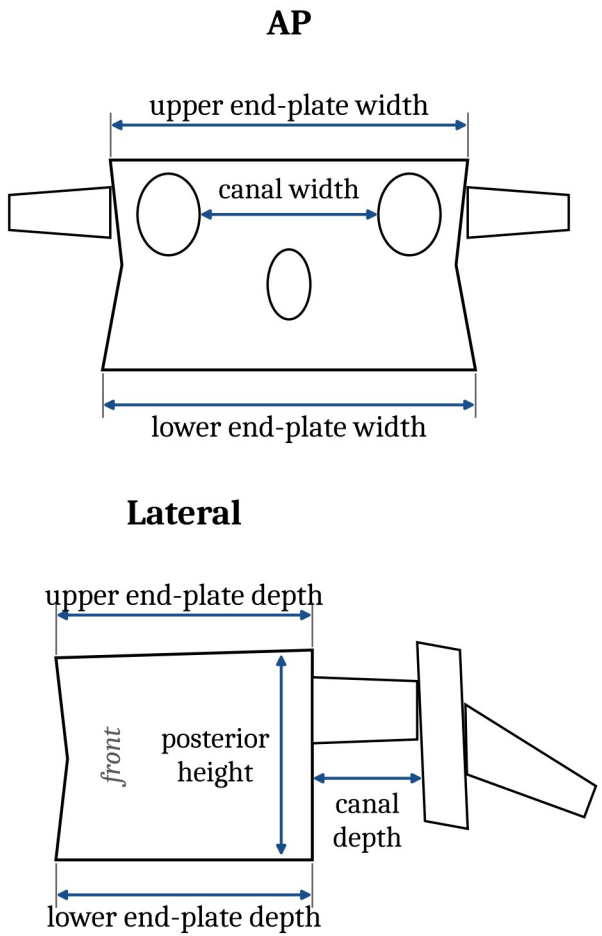
Can the bodies in view name a vertebra?





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A view shows a few consecutive vertebral bodies, often with no C2 or sacrum to count from. Which levels are they?

$$x_{1:n} = (x_1, \ldots, x_n), \quad x_i \in \mathbb{R}^7, \qquad k \in \{1, \ldots, 23 - n\}$$

Seven measures per body. Body i of the field is level $k + i - 1$, with C3 = 1 and L5 = 22, so the place k names every body in it.

2,000 synthetic columns drawn from Panjabi's published means and spreads; every place of every column scored, so most rates below are over tens of thousands of fields.

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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Bayes' rule over every place is the ceiling

$$p(k \mid x_{1:n}) = \frac{p(x_{1:n} \mid k)\, p(k)}{\sum_{k'} p(x_{1:n} \mid k')\, p(k')}$$

posterior ∝ **likelihood** × **prior**

$$p(x_{1:n} \mid k) = \int \prod_{m=1}^{7} \mathcal{N}(x_{m,1:n} \mid s\,\mu_{m,k:k+n-1},\ \Sigma_{m,k:k+n-1}(s))\, p(s)\, ds$$

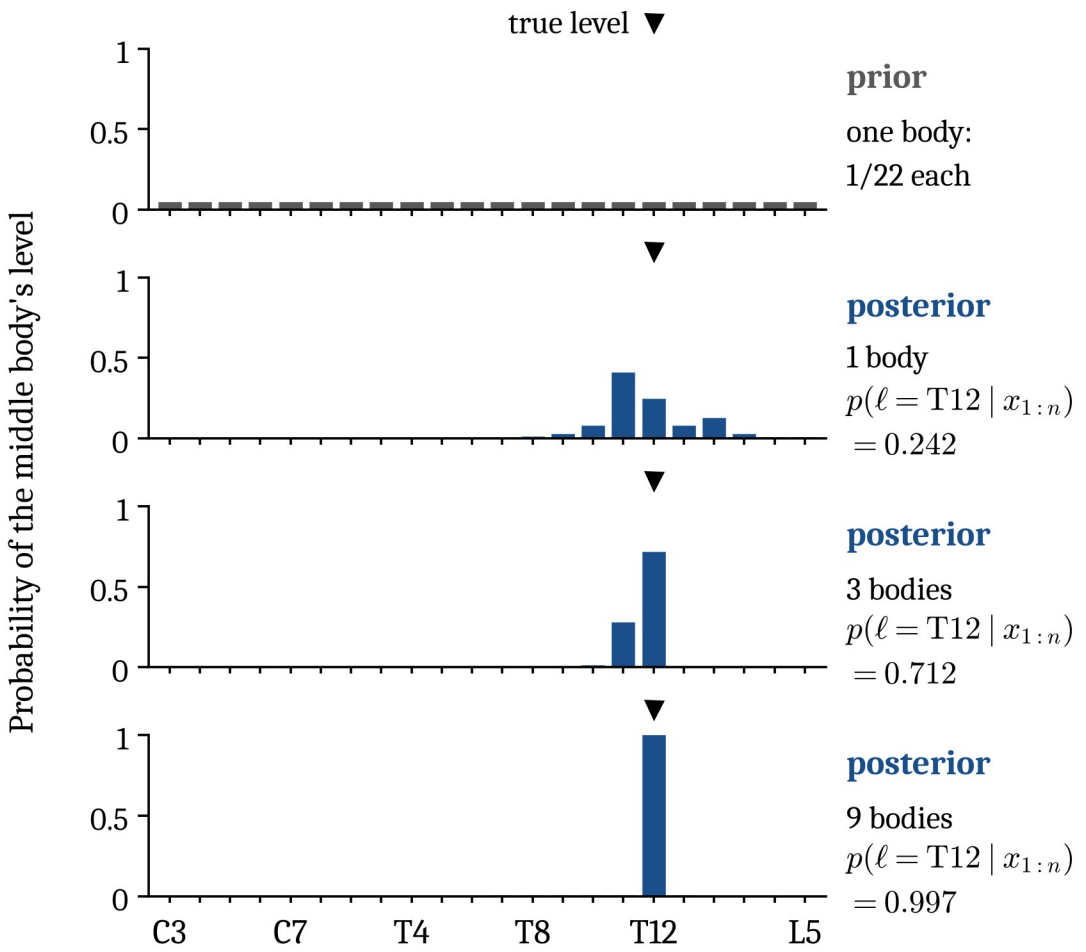
$$\Sigma_{m,k:k+n-1}(s) = s^2 \rho^2\, \sigma\sigma^{\mathrm{T}} + \mathrm{diag}\big(s^2(1-\rho^2)\,\sigma^2 + \sigma_{\mathrm{err}}^2\big)$$

$$\log s \sim \mathcal{N}(0,\ 0.07^2), \qquad \rho = 0.5, \qquad \sigma_{\mathrm{err}} = 1 \text{ mm}$$

Every one of the 23 − *n* places is scored under the prior $p(k) = 1/(23-n)$; nothing is fitted.

σ: the table's spreads at levels *k* to *k* + *n* − 1, scaled by the column's size *s*; the covariance is inverted in closed form and *log s* is integrated by Gauss–Hermite quadrature on 161 nodes.

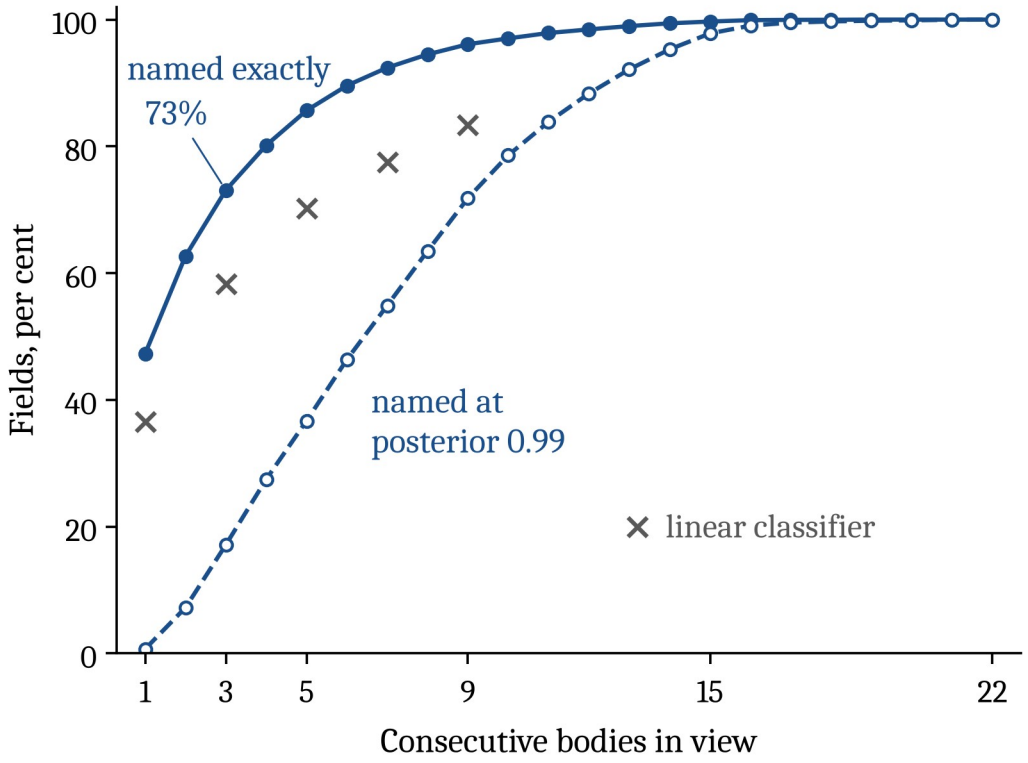
So it is the Bayes rule for this generator: no formula or network on the same measures names more fields correctly on average.

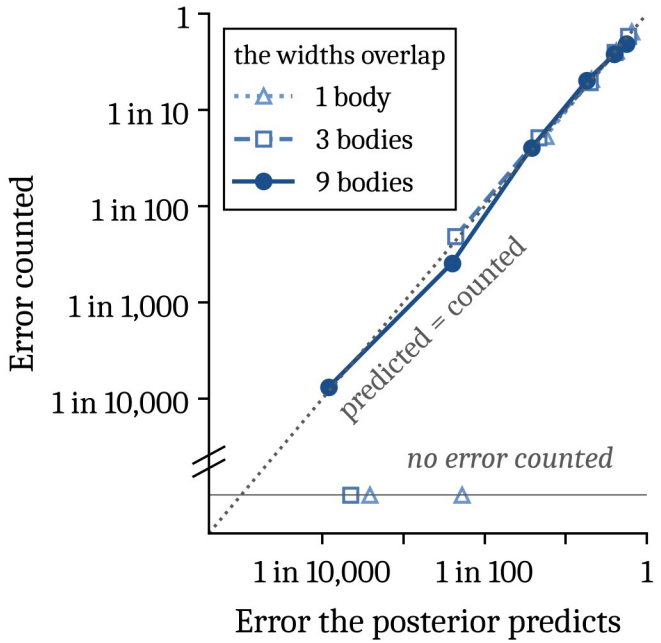


Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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Three bodies name 73%; when sure, 99.8% are right





73%

of three-body fields named exactly (a linear classifier: 58.2%; a fitted network: 71.6%)

99.8%

of three-body names given at posterior 0.99 right (15 wrong of 6,848; named for 17% of fields)

96%

named exactly from nine bodies

Answer only when $\hat{p}$ reaches 0.99, and the name is right at least 99% of the times it is given, in expectation, while the generator is the truth.

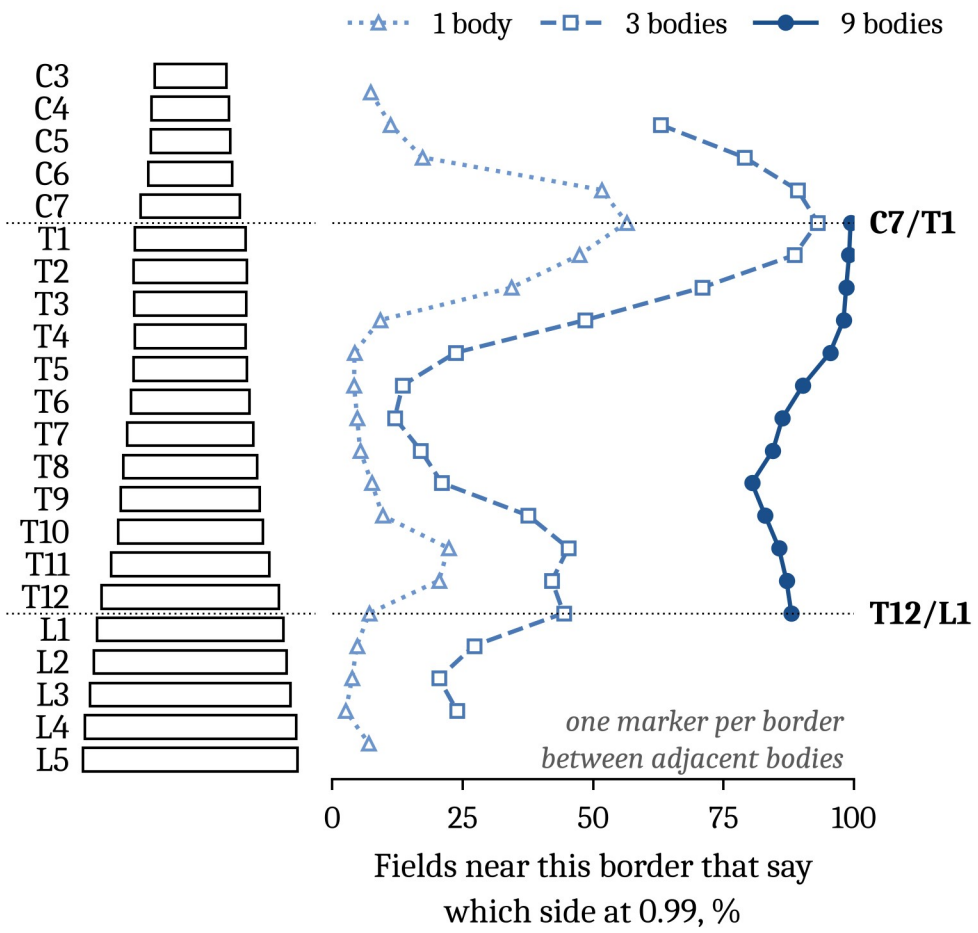
$$\Pr\left(\hat{k} = k \mid \hat{p} \geq 1 - \varepsilon\right) = \mathbb{E}[\hat{p} \mid \hat{p} \geq 1 - \varepsilon] \geq 1 - \varepsilon$$

$$\hat{k} = \arg\max_{k} p(k \mid x_{1:n}), \qquad \hat{p} = p(\hat{k} \mid x_{1:n})$$

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One body tells its region and its side of C7/T1, rarely of T12/L1



$$\Pr(\ell \geq c \mid x_{1:n}) = \sum_{k:\ \ell \geq c} p(k \mid x_{1:n}) \qquad \ell = k + \lfloor (n-1)/2 \rfloor$$

ℓ is the middle body. Claim $\ell \geq c$ when this reaches 0.99 and $\ell < c$ when it falls to 0.01: right at least 99% of the times made, over all fields (T12/L1 from one body: 99.96%, 13 wrong of 30,943). Below, bodies within two levels of a border, a group chosen by the true level, where it need not hold.

56%

of single bodies within two levels of C7/T1 say which side; right 99.8% (7 wrong of 4,514)

7%

of single bodies within two levels of T12/L1 say which side; right **97.9% (12 wrong of 568), short of 99%**

61%

of single bodies say their region; right 99.9% (21 wrong of 26,786), only 2 of them far from both junctions

88%

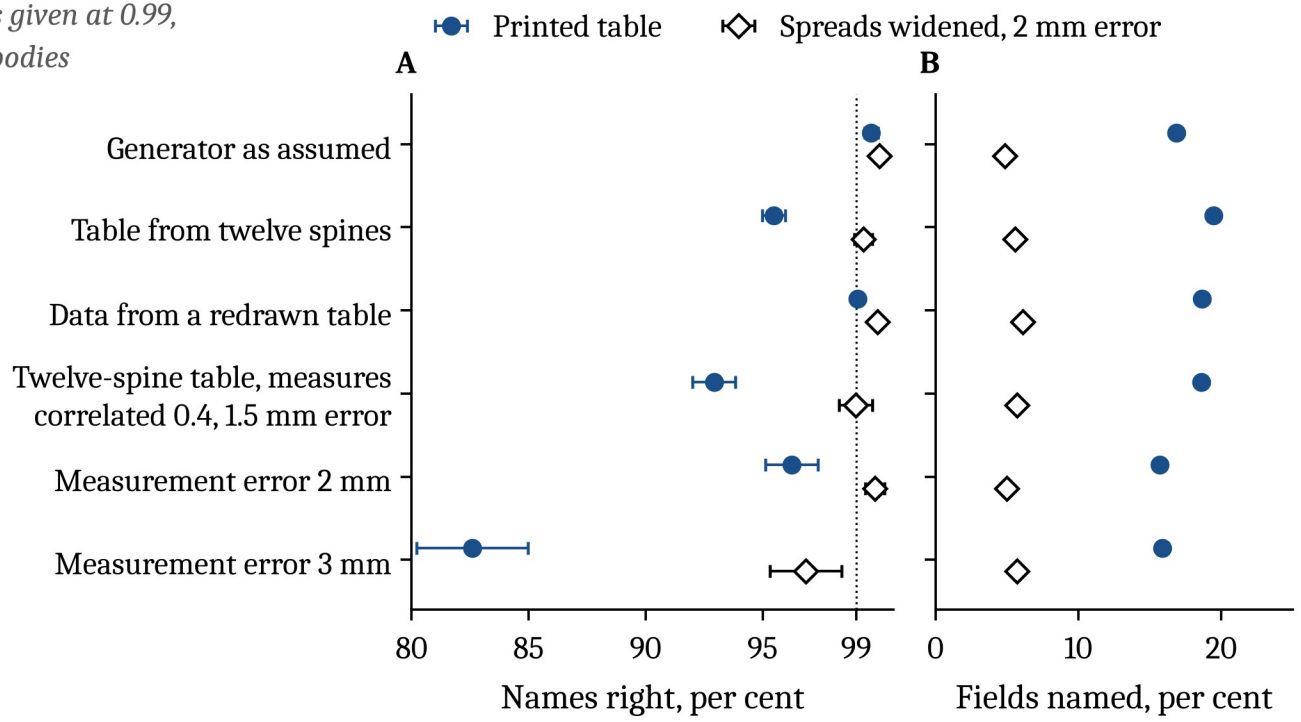
of fields with middle body T11 to L1 say which side of T12/L1 from nine bodies (1 wrong of 5,279)

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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The weak point is the table, not the mathematics

Names given at 0.99, three bodies



95.5%

of three-body names at posterior 0.99 right when the rule's table is estimated from 12 spines, as Panjabi's tables are

99.3%

with spreads widened and 2 mm of error (the 2 mm chosen on a separate pilot), but 5.5% of fields named, not 19.5%, and lumbar names right 96.7% (8 wrong of 246)

$$\tilde{\sigma}_{m,\ell} = \sigma_{m,\ell}\sqrt{1 + 1/12}, \qquad \sigma_{\text{err}} = 2 \text{ mm}$$

Next: measure the table on full-spine CTs counted from C2, and carry a real series' prior over places.

Synthetic columns from Panjabi's published tables (Spine 1991;16:861, 888; 1992;17:299); no patient data.

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